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Anand Nagaraj

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EDUCATION AND TRAINING

[01/04/2026 – Current]

Master of Science in Design and Test of Integrated Circuits (DTIC)

Technische Universität Chemnitz

City: Chemnitz | **Country:** Germany | **Level in EQF:** EQF level 7

• Core Coursework: Verification of Digital Systems, Integrated Circuit Design (Transistor Level), Design of Heterogeneous Systems, Advanced Integrated Circuit Technology, Technologies of Micro/Nano Systems.

• Practical chip verification, digital architecture modeling, and transistor-level laboratory components.

• Academic Projects: Architected a dual-clock digital measurement unit in VHDL with cooperating FSMs and request-acknowledge CDC handshakes; modeled continuous-time sensor front-ends in VHDL-AMS.

• Honors: Recipient of the Deutschlandstipendium scholarship (2026/2027) sponsored by Infineon Technologies Dresden.

[01/08/2018 – 30/04/2022]

Bachelor of Engineering in Electronics & Telecommunication Engineering

Visvesvaraya Technological University (VTU / BMSIT)

City: Bengaluru | **Country:** India | **Level in EQF:** EQF level 6

• Core Coursework: CMOS VLSI Design, Computer Architecture, Digital System Design, Microprocessors, Signals and Systems.

• Honors & Awards: Conferred Best Outgoing Student in the Department; Recipient of Karnataka State Merit Scholarship (100% tuition waiver).

• Academic Standing: Graduated with CGPA 8.51/10.0 (German equivalent GPA 1.8).

• Research & Projects: Conducted capstone research on PCIe interfaces, later published as "Peripheral Shared Memory Over PCI Express Link" at IEEE ICIDCA 2025.

WORK EXPERIENCE

IndusIP Labs Pvt. Ltd

City: Bengaluru | **Country:** India

[01/11/2024 – 28/02/2026]

VLSI Design & Verification Engineer (Founding Team)

• Worked extensively with the open-source RISC-V Ibex core, integrating custom instruction extensions and dedicated hardware execution logic for specialized workloads; validated designs via FPGA prototyping.

• Designed synthesizable SystemVerilog RTL for 2D-mesh Network-on-Chip (NoC) routers, implementing virtual channel allocators, round-robin arbiters, and crossbars.

• Implemented SystemVerilog Assertions (SVA) across asynchronous FIFO clock-domain crossings (CDC) to eliminate transaction deadlocks under heavy synthetic traffic.

• Built simulation testbenches in SystemVerilog and Cocotb; developed automated Python regression frameworks to aggregate simulation logs and track functional coverage.

Sinergia Media Labs (Simelabs)

City: Bengaluru | **Country:** India

[01/09/2022 – 21/10/2024]

RTL Design & Verification Engineer

• Designed synthesizable Verilog RTL for high-throughput PCIe Transaction Layer IP, implementing header parsing, packet framing, and flow control buffer management.

• Implemented end-to-end data integrity protection using 32-bit CRC (ECRC) generation/checking and poisoned packet detection logic.

• Developed modular UVM testbenches and Python Cocotb co-simulation harnesses to validate transaction ordering and protocol corner cases.

• Synthesized RTL blocks with Synopsys Design Compiler, verifying setup/hold margins and closing static timing constraints.

SKILLS

Hardware Description Language: Verilog | System verilog | RISC V | Python | C++ | C | Linux | VHDL | Git | Knowledge of PCIe | Communication protocols: ZigBee, Bluetooth, USB FS, SPI, I2C, UART, SPI, AMBA, etc | FPGA(VHDL - basic experience with Xilinx ISE) | Constrained Random UVM Test-benches | Universal Verification Methodology (UVM) | cocotb | Shell (Bash, Tcsh)

PROJECTS

[01/08/2026 – Current]

Moola-V (MR5) Processor

Project: Moola-V (MR5) - 5-Stage Pipelined RV32IM Processor Core (SystemVerilog / Cocotb)

• Architected an in-order, 5-stage pipelined RV32IM core in synthesizable SystemVerilog featuring complete RAW data hazard forwarding (EX-to-EX, MEM-to-EX) and cycle-accurate load-use stall mechanisms.

• Integrated dedicated datapath execution units, including a single-cycle multiplier and a 32-cycle iterative radix-2 restoring divider.

• Developed a custom Python Cocotb co-simulation test harness against the Spike Golden Instruction Set Simulator (ISS).

• Verified 100% architectural compliance across all 49 RISCOF architectural test suites.

Link: https://github.com/AnandNag003/moola_riscv_core

[01/09/2026 – 31/10/2026]

Dual-Port Synchronous SRAM

Project: Parameterized Dual-Port Synchronous SRAM Macro (SkyWater 130nm / OpenLane 2)

• Designed and verified a parameterized 16x8 dual-port synchronous SRAM macro module featuring independent write-masking and concurrent read/write arbitration logic within the open-source Tiny Tapeout framework.

• Executed the complete physical implementation flow from synthesizable RTL down to GDSII using OpenLane 2 and OpenROAD.

• Achieved clean DRC/LVS physical verification and closed static timing analysis (STA) sign-off on the SkyWater 130nm open-source PDK.

Link: https://github.com/AnandNag003/tt_moola_dp_sram

[01/05/2022 – 31/08/2022]

AMBA AHB-to-APB Protocol Bridge & Verification Suite

Project: AMBA AHB-to-APB Protocol Bridge & Verification Suite (Verilog / SystemVerilog)

• Designed synthesizable RTL for an AMBA-compliant AHB-to-APB bridge, implementing cycle-accurate FSM control logic to translate pipelined AHB transfers into multi-phase APB bus cycles.

• Implemented address decoding, wait-state propagation via HREADYOUT, error response mapping, and transient register buffering.

• Developed a self-checking SystemVerilog testbench featuring SystemVerilog Assertions (SVA) and functional coverage models to verify burst sequences, transfer timeouts, and protocol corner cases.

[01/05/2026 – 31/07/2026] **Mixed-Signal Frequency Measurement**

Project: Mixed-Signal Frequency Measurement & Interface System (VHDL / VHDL-AMS)

• Architected a reliable dual-clock digital measurement unit in VHDL, implementing cooperating FSMs across asynchronous domains with double-flop synchronizers and request-acknowledge CDC handshake protocols.

• Modeled a continuous-time MEMS analog front-end (sensor, RC low-pass filter, and VCO) in VHDL-AMS.

• Integrated the analog behavioral model with digital threshold decision logic and a 115200-baud UART transmitter for end-to-end system verification.

PUBLICATIONS

[2025] **Peripheral Shared Memory Over PCI Express Link**

Reference: A. Nagaraj et al., "Peripheral Shared Memory Over PCI Express Link," in Proceedings of the 7th IEEE International Conference on Innovative Data Communication Technologies and Application (ICIDCA 2025).

Analyzed shared-memory throughput, buffer sizing trade-offs, and transaction latency across high-speed PCIe interfaces. Evaluated transaction layer packet processing and memory mapping performance for peripheral-to-host and peripheral-to-peripheral interconnects.

Authors: A. Nagaraj et al | **Journal Name**: 7th International Conference on Innovative Data Communication Technologies and Application (ICIDCA) | **Publisher**: IEEE

Link: https://ieeexplore.ieee.org/document/11280455

CERTIFICATIONS

[Maven Silicon, 30/09/2022] **Certificate in Advanced VLSI Design and Verification**

• 6-Month Full-Time Vocational Program specializing in digital VLSI design and advanced functional verification.

• SystemVerilog & UVM: Architectured modular, reusable UVM testbench components (drivers, monitors, scoreboards, sequencers).

• On-Chip Protocols: In-depth design and verification coverage of AMBA protocols (AXI, AHB, APB).

• Verification Methodologies: Implemented SystemVerilog Assertions (SVA), functional coverage models, and constrained-random test sequences.

• EDA Tools: Hands-on simulation, waveform debugging, and coverage analysis using Siemens QuestaSim.

LANGUAGE SKILLS

Other language(s):

English

LISTENING C2 **READING** C2 **WRITING** C2

SPOKEN PRODUCTION B2 **SPOKEN INTERACTION** B2

German

LISTENING A2 **READING** A2 **WRITING** B2

SPOKEN PRODUCTION A2 **SPOKEN INTERACTION** A2

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

HONOURS AND AWARDS

[01/10/2026] **Deutschlandstipendium National Scholarship (2026/2027) Awarding institution:** Technische Universität Chemnitz & Infineon Technologies Dresden

Awarded the prestigious German national merit scholarship (Deutschlandstipendium) for the 2026/27 academic year at TU Chemnitz, sponsored by Infineon Technologies Dresden. Granted in recognition of outstanding academic performance, leadership potential, and technical commitment to semiconductor technologies.

[01/08/2018] **Karnataka State Merit Scholarship Awarding institution:** Karnataka Board of Education / Government of Karnataka

Conferred a 100% full undergraduate tuition waiver throughout the Bachelor of Engineering program for ranking among the top candidates in the state competitive entrance examination.

[30/04/2022] **Best Outgoing Student Award (Batch of 2022) Awarding institution:** Visvesvaraya Technological University (VTU / BMSIT)

Conferred by the Department of Electronics & Telecommunication Engineering for sustained academic excellence (German equivalent GPA 1.8 / CGPA 8.51/10.0), capstone research contributions, and leadership.

ACHIEVEMENTS

[10/11/2026 – 12/11/2026] **Participant in SEMICON Europa × TUM Venture Labs Hackathon (NextSEMI 2026)**

Selected as one of only 25 participants across Europe for an intensive semiconductor and hardware prototyping hackathon.

HOBBIES AND INTERESTS

Sports & Outdoor Activities

• Competitive Sports: 2nd Place in District-level Kabaddi Championship (Under-19).

• Distance Running & Fitness: Active 5 km road runner; regular progressive resistance strength training.

• Outdoor Recreation: Regional day-hike enthusiast across Saxony (Fichtelberg, Saxon Switzerland).

• Team Sports: Cricket.

VOLUNTEERING

[01/05/2026 – Current] **Student Volunteer — IEEE Youth Wing Chemnitz** Chemnitz, Germany

• Actively support the IEEE Student Branch and Youth Wing at TU Chemnitz in organizing technical workshops, semiconductor seminars, and student networking initiatives.

• Assist international engineering students with academic integration and technical event coordination.

[01/08/2019 – 30/04/2022] **Academic Symposium Coordinator & Civic Volunteer** Bengaluru, India

• Coordinated departmental technical symposiums and cultural events during undergraduate studies at VTU/BMSIT.

• Volunteered during regional elections, assisting elderly and disabled voters with polling administration and accessibility logistics.